

texforge

A self-healing $\text{\LaTeX} \rightarrow \text{PDF}$ engine

Rendered by the texforge engine on tectonic

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Abstract

This report is a fixture: it exercises sectioning, cross-references, a table, an equation and a bulleted list so a single build proves the toolchain end to end. Because tectonic fetches missing packages on demand, this file compiles from a clean machine with no $\text{\TeX}$ installation.

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1 Introduction

texforge wraps the tectonic engine with an observable CLI. It streams structured logs, times every build, and reads the engine’s own output to suggest fixes for common problems — see Section 3.

2 A little mathematics

The Gaussian integral, for good measure:

$$\int_{-\infty}^{\infty} e^{-x^2} \mathrm{d}x = \sqrt{\pi}. \tag{1}$$

Equation (1) typesets cleanly with `amsmath`.

3 Results

A small table via `booktabs`:

Design goals:

- **Self-healing** — packages arrive on demand.
- **Self-optimizing** — caching plus log-driven suggestions.
- **Observable** — level-controlled, structured logs.
- **Portable** — one static binary per OS, no runtime deps.

Stage	Cold build	Warm build
Fetch packages	yes	cached
Compile passes	2	2
Total	slower	fast

Table 1: Cold vs. warm builds. Warm builds reuse the package cache.

4 Conclusion

If you can read this PDF, the engine works. See the project [README](#) for the quickstart.